The Inflation Rate is the rate at which the circulating supply of a cryptocurrency is increasing over time, **expressed as a percentage of the total supply**. This rate is calculated as the *difference between the circulating supply and the total maximum supply (21 mill), divided by the total maximum supply and multiplied by 100.*

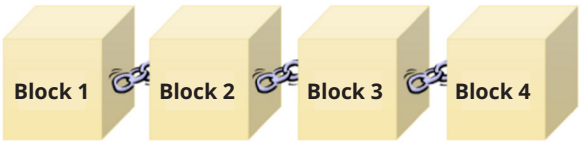
During each **halving event**, the block reward for miners is reduced, lowering the **issuance rate** of new ***bitcoin***. As a result, the **inflation rate** of ***bitcoin*** decreases over time, potentially leading to an increase in the price of ***bitcoin***.

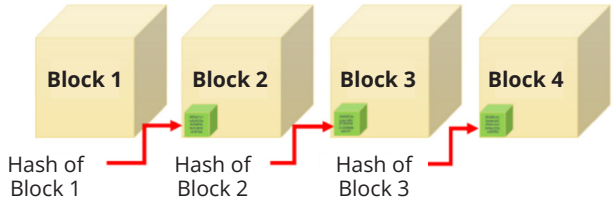
The **reduced supply**, combined with **increasing demand**, can drive up the price of ***bitcoin***. This not only benefits early adopters of the technology, but it also serves as an incentive for miners to continue to secure the network and contribute their computing power and resources.

8.2 The Vital Task of Bitcoin Mining: Securing the Blockchain

What is a Valid Block Hash in Blockchain?

A **block hash** serves as a **unique identifier** for each block in the *blockchain* and helps detect any attempts to alter past transactions. The **blocks in the blockchain store transactions and form a chain of blocks**, starting from the **genesis block** to the most recent one, creating a public and transparent record of all transactions. The block hash links each block to the previous one, allowing anyone to view the history of any transaction and ensuring the accuracy and security of the data stored on the network. Just like a **fingerprint** identifies an individual, the block hash identifies each unique block in the *blockchain*.



The blocks are "linked" together by enforcing a specific relationship between blocks. That is, a block must contain a "fingerprint", which is a hash value of the data of the previous block. A hash function can condense arbitrary message (the block information) to a fixed size (e.g., 160 bits) and produces a finger print of the message.

The first ever block of ***bitcoin***, containing total of 50 ***bitcoin*** was mined by the creator of ***bitcoin***, Satoshi Nakamoto.